

$$\begin{array}{ccc}
\iota A & \xrightarrow{\quad \iota \operatorname{coev}_A^B \quad} & \iota[B, A \otimes_{\mathcal{D}} B]_{\mathcal{D}} \\
\downarrow \operatorname{coev}_{\iota A}^{\iota B} & & \uparrow \psi_{B, A \otimes_{\mathcal{D}} B}^{-1} \\
[\iota B, \iota A \otimes_{\mathcal{C}} \iota B]_{\mathcal{C}} & \xrightarrow{\quad [\iota B, \eta_{\iota A \otimes_{\mathcal{C}} \iota B}] \quad} & [\iota B, \iota(A \otimes_{\mathcal{D}} B)]_{\mathcal{C}}
\end{array}$$